Arm Cortex-A78 Technical Reference Manual - DATASHEET

Manufacturer: ARM
Part Number: Cortex-A78
Category: CPU / Mobile CPU Core
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Overview

The Arm Cortex-A78 is a high-performance CPU core implementing the Armv8.2-A architecture, licensed as IP for mobile, laptop and embedded SoCs. It delivers about 20% sustained performance uplift over Cortex-A77 at the same power. 4-wide decode, out-of-order pipeline, optional Neon SIMD and cryptography extensions. Status: Not Recommended for New Designs; Arm recommends Armv9 cores for new projects.

Key Specifications Summary

arch: ARMv8.2A. pipeline: 13stage. ooo: yes. neon: 128bit. cache_L1: 64KB. cache_L2: 512KB.